PARACETAMOL

Pregnancy: analgesic of choice

Hepatotoxic in overdose

CHEMICAL NAME N-(4-hydroxyphenyl)acetamide
CLASS Non-opioid analgesic and antipyretic (para-aminophenol derivative)
TRADE NAMES Panadol · Tylenol · Adol · Calpol · Perfalgan (IV)

THERAPEUTIC USES / INDICATIONS

- Mild to moderate pain: headache, toothache, musculoskeletal pain, dysmenorrhoea, post-operative pain.
- Fever in adults and children.
- Part of multimodal analgesia with NSAIDs or opioids (opioid-sparing).

ADMINISTRATION

- **PO:** tablets, syrup, drops; with or without food.
- **IV:** infuse over **15 min**; do not give as a bolus.
- **PR:** suppositories when oral route is not possible.
- Keep **at least 4 h** between doses; record every dose given.

DOSAGES

- **Adult PO:** 0.5–1 g every 4–6 h; max 4 g/day.
- **Adult IV (>50 kg):** 1 g every 6 h; max 4 g/day.
- **IV ≤50 kg / children:** 15 mg/kg every 6 h; max 60 mg/kg/day.
- **Children PO:** 10–15 mg/kg every 4–6 h; max 4 doses/24 h.
- **Liver risk** (alcohol, malnutrition, low weight): max 2–3 g/day.

PHARMACOKINETICS

- **Onset:** PO 30–60 min; IV 5–10 min. **Duration:** 4–6 h.
- Well absorbed orally; metabolised in the liver (glucuronidation, sulfation); a small amount forms toxic NAPQI, neutralised by glutathione.
- Excreted in urine. **t½:** 2–3 h.

PHARMACODYNAMICS

- Inhibits cyclo-oxygenase (COX) mainly in the **CNS** → ↓ prostaglandins → analgesia.
- Acts on the hypothalamic heat-regulating centre → antipyretic.
- Weak peripheral COX inhibition → no useful anti-inflammatory effect; no gastric or platelet effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Rare at normal doses; nausea.
- Skin rash.
- IV: transient hypotension, infusion-site pain.

ADVERSE EFFECTS (SERIOUS)

- **Overdose:** hepatotoxicity — nausea and vomiting first, liver failure at 48–72 h. Antidote: N-acetylcysteine.
- Thrombocytopenia (rare).
- SJS/TEN (very rare).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to paracetamol.
- Severe hepatic impairment or active liver disease.
- **Caution:** chronic alcohol use, G6PD deficiency, dehydration, weight <50 kg.

KEY INTERACTIONS

- **Warfarin:** regular use raises INR.
- **Enzyme inducers** (carbamazepine, phenytoin, rifampicin), **alcohol:** ↑ liver toxicity.

NURSING MANAGEMENT

- Assess pain score or temperature before giving; reassess 30–60 min after and document.
- Check all medicines given in the last 24 h for hidden paracetamol (combination products).
- Weigh children and adults <50 kg; calculate mg/kg dose and double-check.
- IV: give over 15 min; monitor BP; record the time to keep the 4–6 h interval.
- Long-term or high-dose use: monitor LFTs.

PATIENT EDUCATION

- Never exceed the daily maximum; many cold and flu products contain paracetamol.
- Avoid alcohol.
- After an overdose seek help immediately, even without symptoms.
- Use the measuring syringe supplied for children's syrup.

KEY POINTS

- Analgesic + antipyretic, **not** anti-inflammatory.
- Overdose → liver failure; antidote N-acetylcysteine.
- Adult max 4 g/day; lower in liver risk.
- Preferred analgesic in pregnancy.

SOURCES CHECKED [1] Paracetamol 10 mg/ml solution for infusion (UK SmPC) [2] Acetaminophen, IV (US label)

Full references p. 432–457